

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

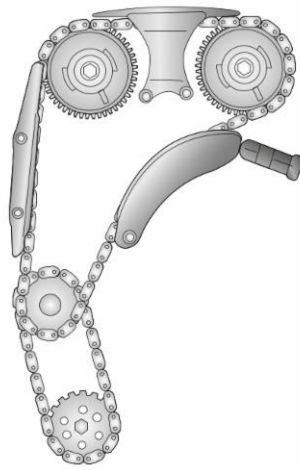
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

Always use new gaskets

General

Timing chain layout



Removal

Remove the cowl lower grille

Disconnect the battery earth cable

Remove the engine

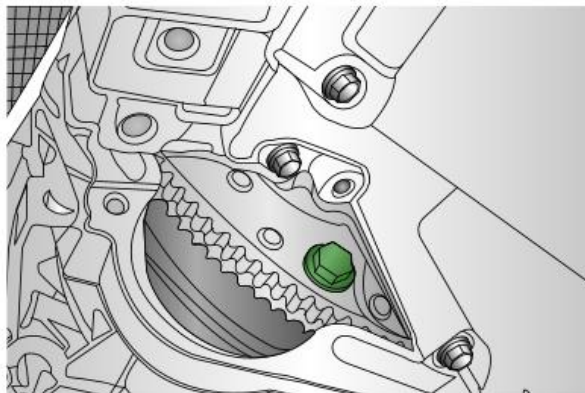
Remove the transfer box

Remove the gearbox lower cover

Fit the crankshaft turning tool (11 6 480)

Turn the crankshaft clockwise

Remove the torque converter attaching bolts



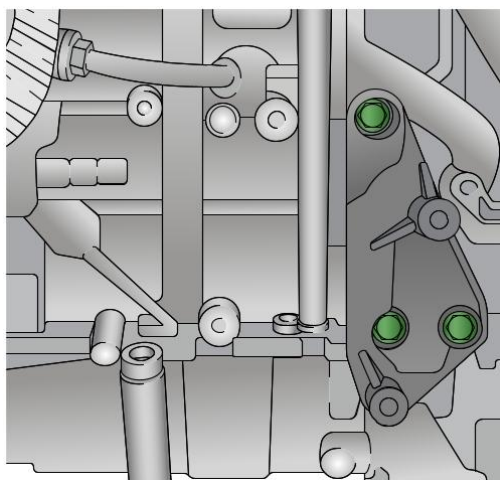
Remove the engine/gearbox attaching bolts

Separate the gearbox from the engine

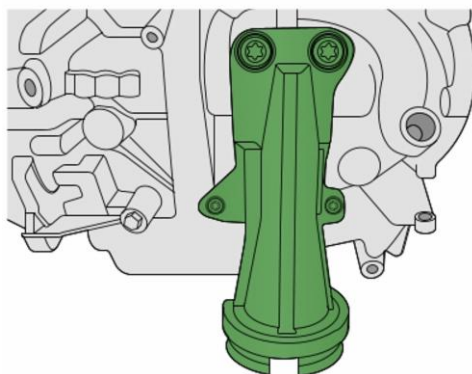
Fit a lifting eye to the cylinder head ((21.5 Nm)) (2 459 012)

Note: Lift the engine (11 0 070)

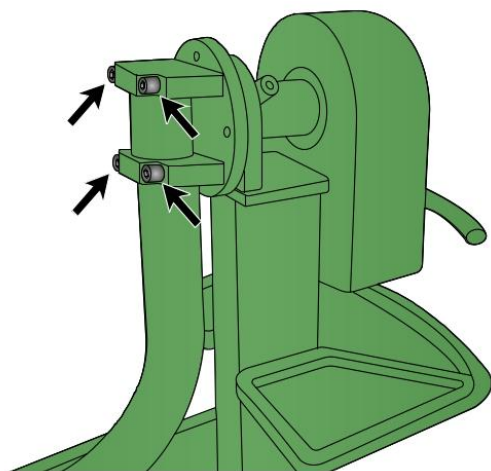
Remove the engine support bracket

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Fit the special tool ((38 Nm)) (2 455 010)

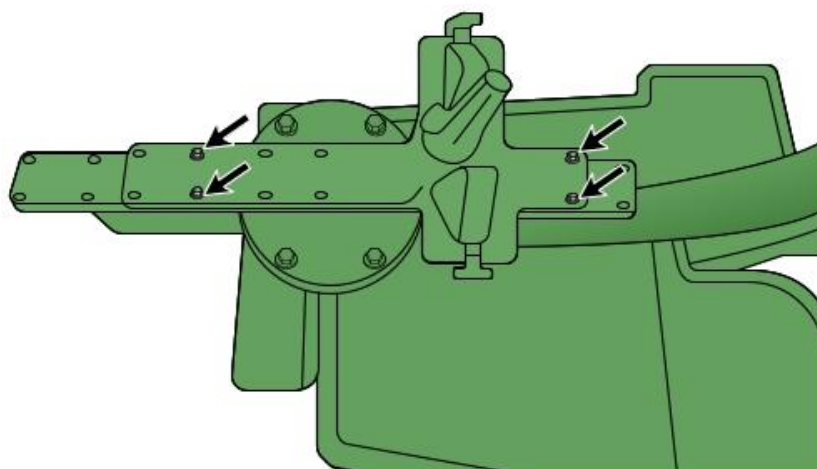


Assemble the special tool as shown. Tighten the special tool bolts ((130 Nm)).



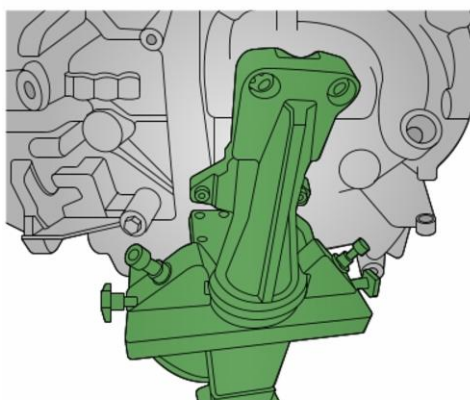
Fit the special tool ((160 Nm)) (2 413 082)

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Lower the engine carefully

Tighten the special tool (2 413 082, 2 455 010)



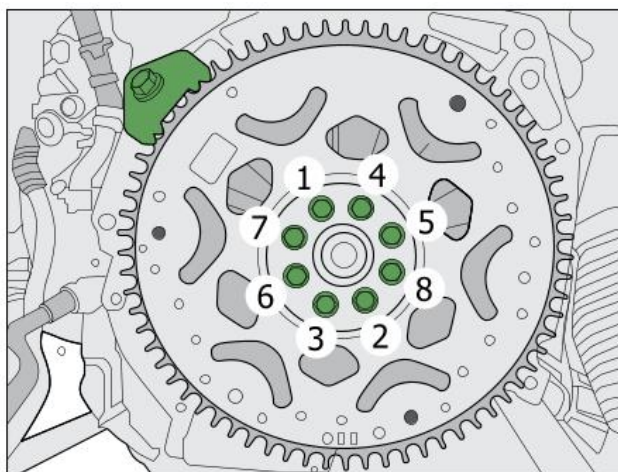
Remove the cylinder head

Fit the flexplate locking tool (11 9 811)

With washer: (5 mm)

Secure the special tool with a bolt (11 9 265)

Remove the bolts in the order shown ((1 - 8))

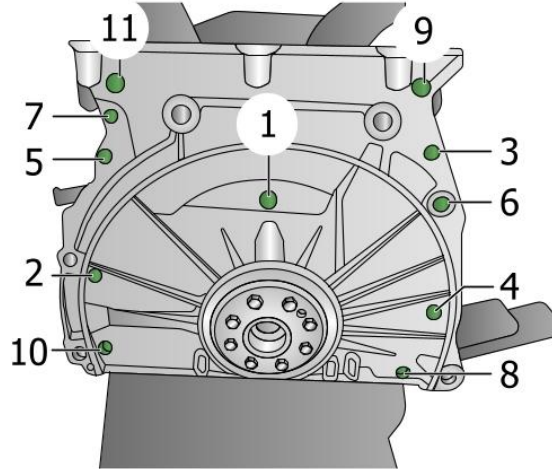


Remove the flexplate

Remove the crankshaft position sensor plate

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- Disconnect the oil level sensor
- Disconnect the wiring loom from the oil sump
- Remove the lower torque reaction rod
- Remove the oil sump
- Remove the starter motor
- Disconnect the crankshaft position sensor
- Remove the bolts in the reverse order to that shown ((11 -> 1))



- Remove the timing chain cover
- Remove the chain tensioner guide
- Remove the chain guide
- Remove the timing chain
- Remove the oil pump gearwheel
- Remove the oil pump chain

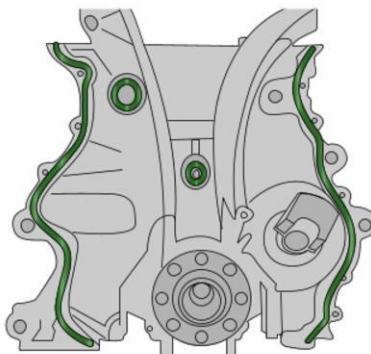
Installation

- Installation is in the reverse order of removal
- Note: Clean off the sealing surfaces

Timing chain cover:

- Remove and discard the crankshaft oil seal
- Apply sealant as indicated ((2.0 - 2.5 mm))

Note: The cover must be fitted within 10 minutes of applying the sealant



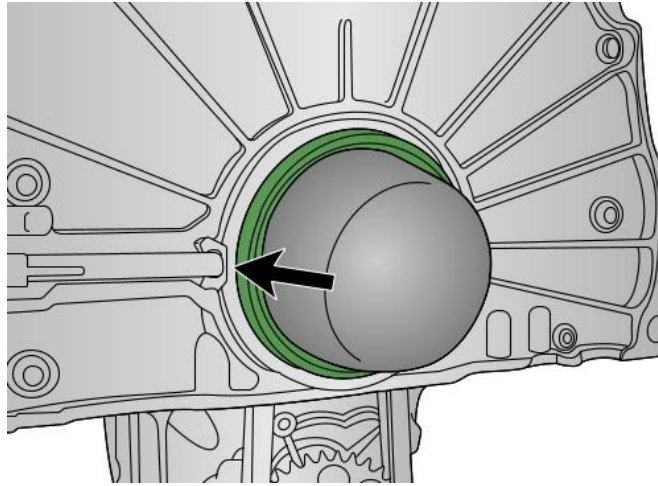
- Refit the timing cover
- Reconnect the crankshaft position sensor
- Fit the special tool (11 8 815)

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Note: Do not apply lubricant

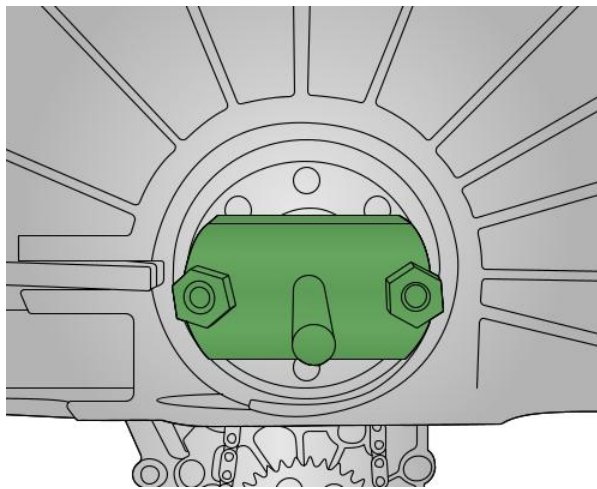
Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal

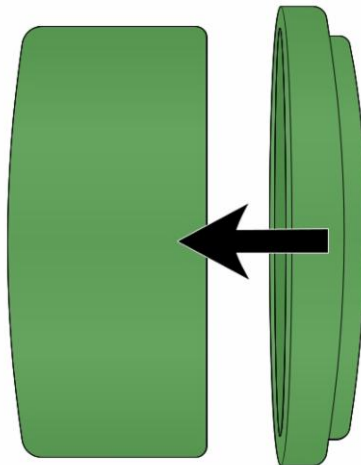


Remove the special tool (11 8 815)

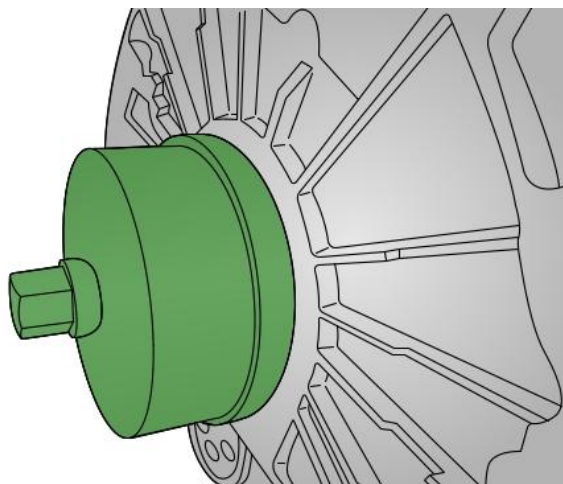
Fit the special tools (11 9 181, 11 9 182, 11 9 183, 11 9 184, 11 9 185)



Assemble the special tool as shown (2 360 215)



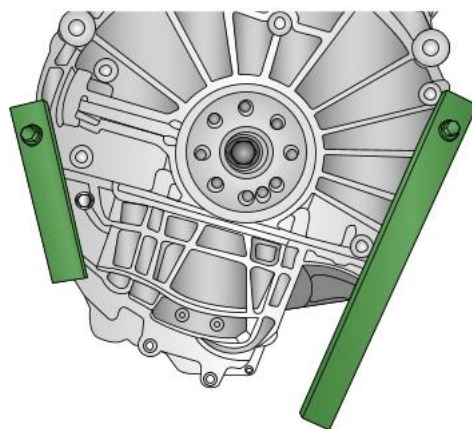
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Tighten the special tool bolt until full surface contact is made

Refit the oil sump

Use the special tools (11 8 780)



Refit the lower torque reaction rod

Refit the crankshaft position sensor plate

Refit the flexplate

Further refitting is in the reverse order of removal

Renew the gearbox output seals

Reconnect the battery earth cable

Torque settings

Oil pump gearwheel

Oil pump gearwheel (Left-hand thread)

Renew the bolt

Stage 1 5 (Nm)

Stage 2 90 (°)

Timing chain tensioner guide

Timing chain tensioner guide 20 (Nm)

Renew the bolts

Timing chain guide

Timing chain guide (To crankcase) 20 (Nm)

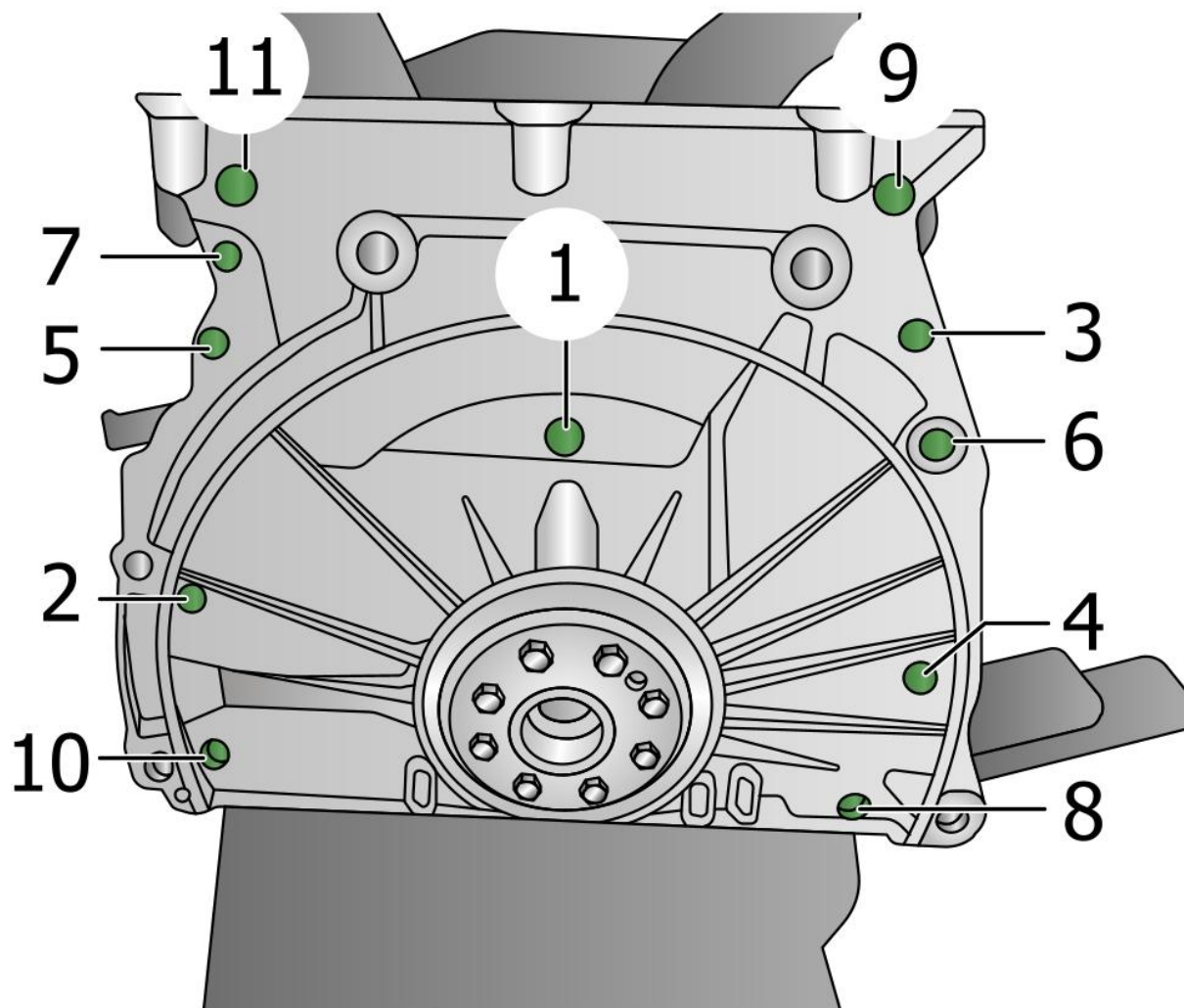
Renew the bolts

Timing chain guide (To cylinder head) 22 (Nm)

Timing cover

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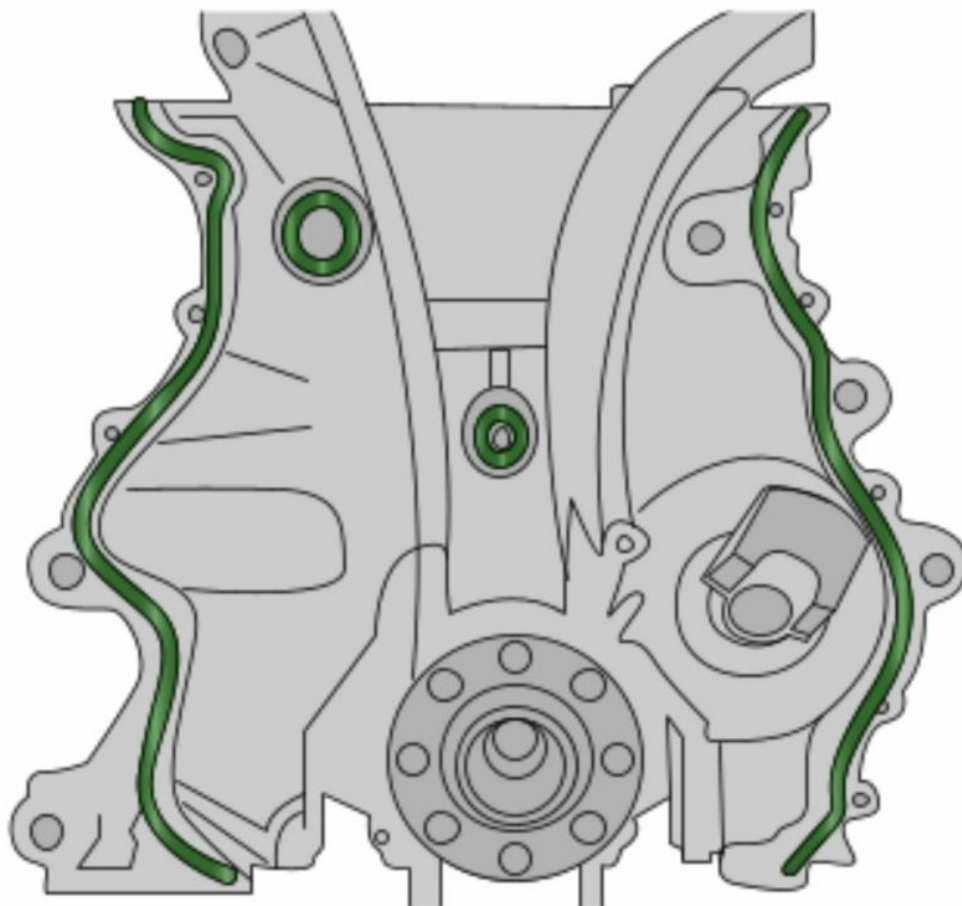
Timing cover



Renew the bolts

Clean the contact surfaces

Apply sealant 2.0 - 2.5 (mm)

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The component must be fitted within 10 minutes of applying the sealant

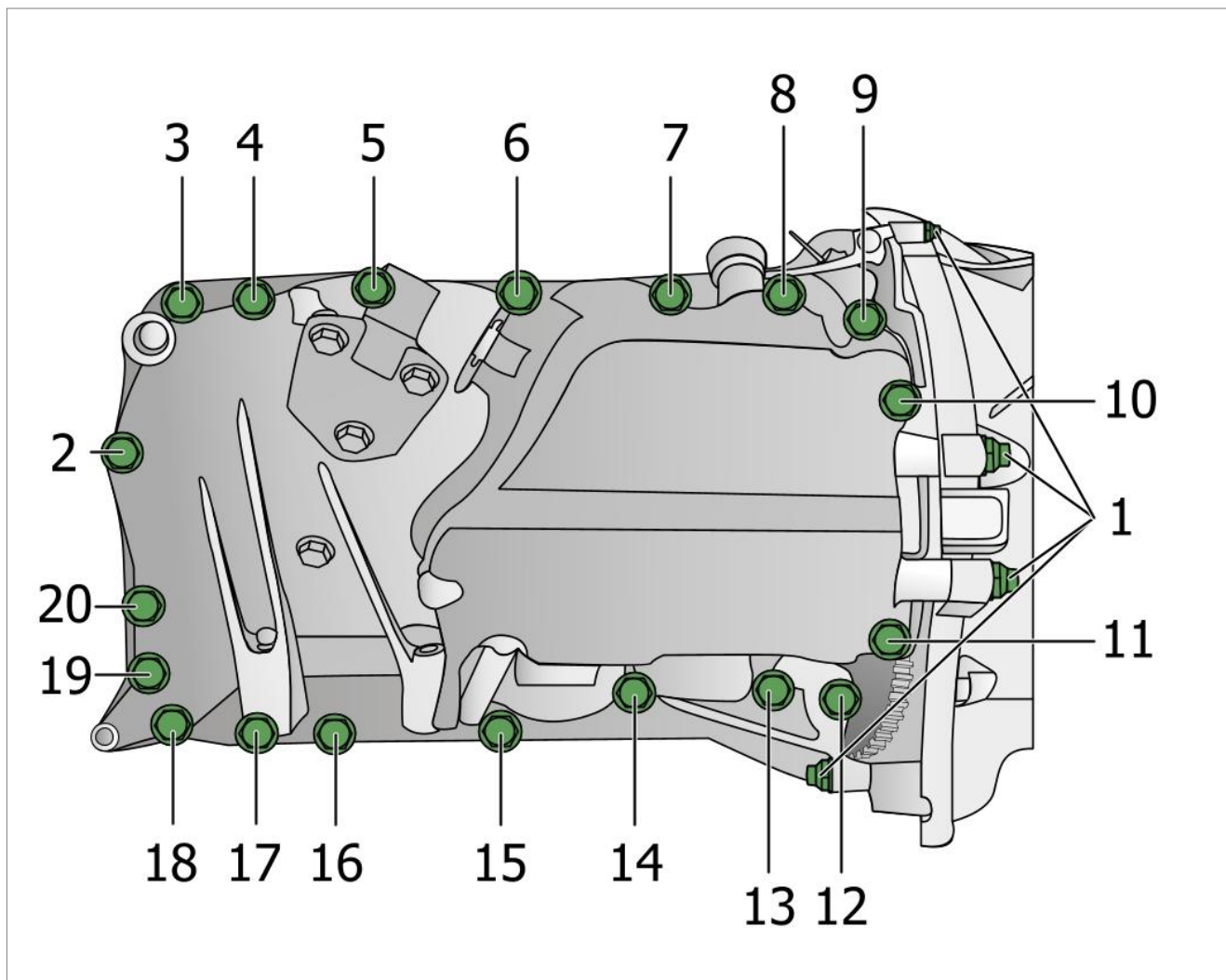
Wait for 25 minutes before starting the engine

Stage 1 8 (Nm)

Stage 2 90 (°)

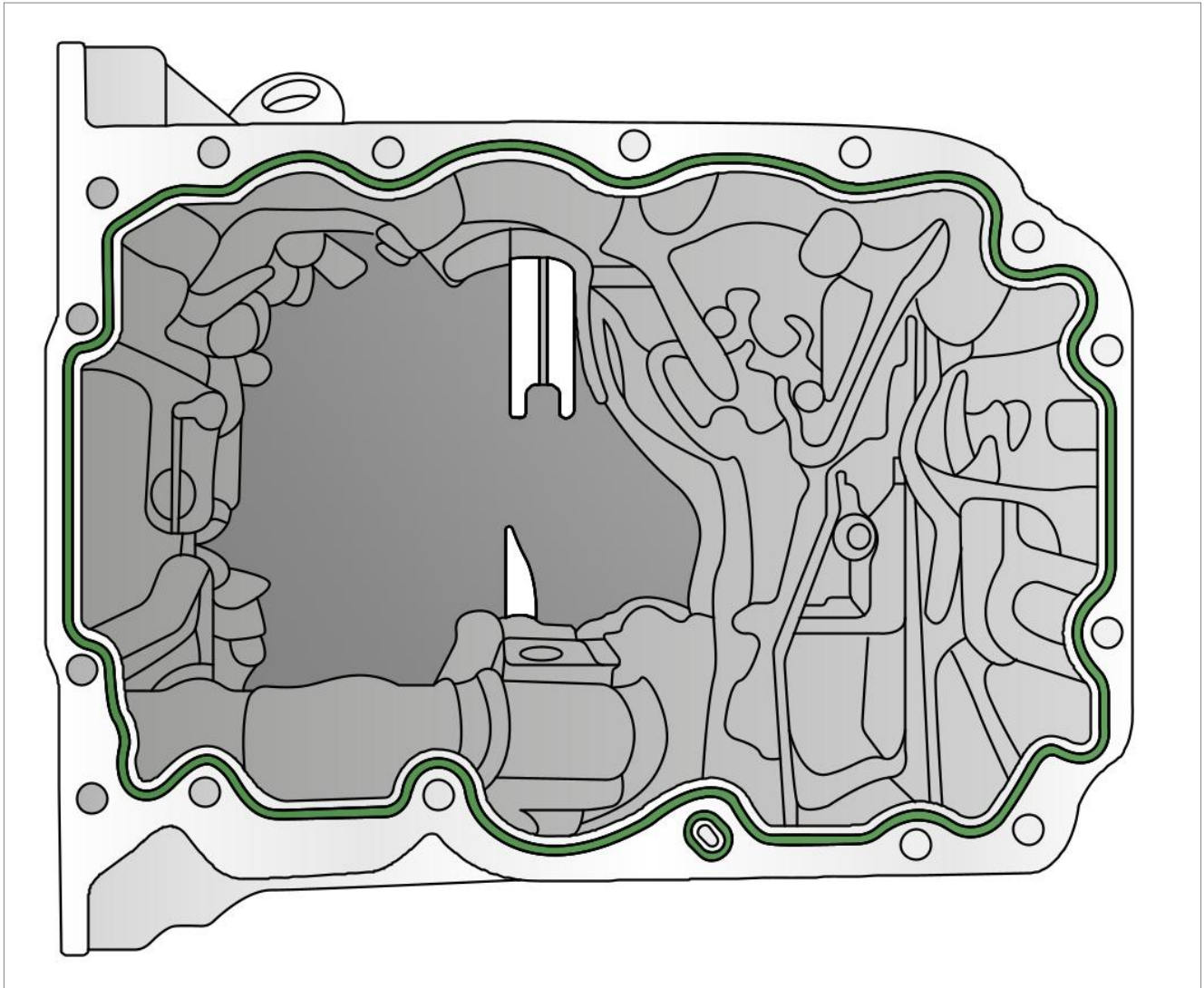
Oil sump

Sump

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Clean the contact surfaces

Apply sealant 2.0 - 2.5 (mm)

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The component must be fitted within 10 minutes of applying the sealant

Wait for 25 minutes before starting the engine

Stage 1 (Bolts 2 - 20) 24 (Nm)

Stage 2 (Bolts 1) 19 (Nm)

Flexplate

Flexplate

Renew the bolts

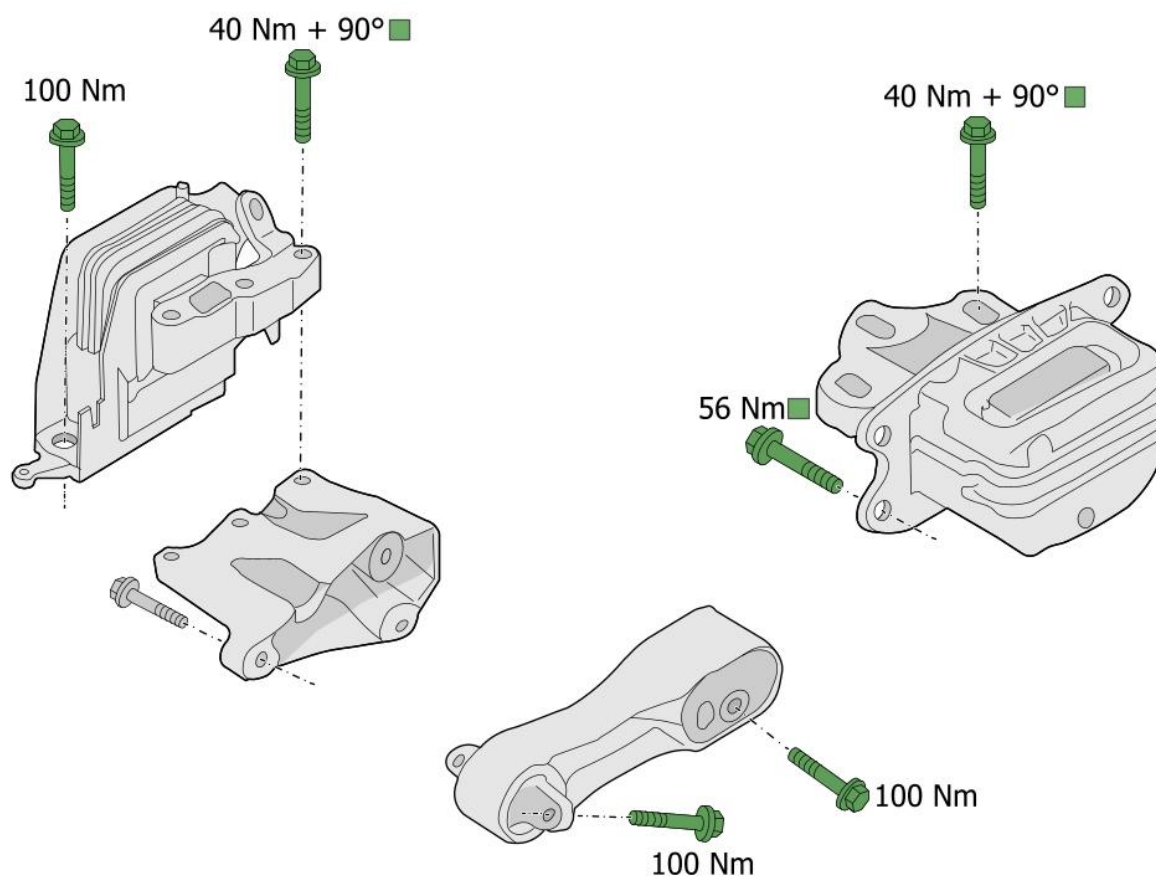
Stage 1 45 (Nm)

Stage 2 45 (°)

Engine/gearbox mounts

Engine

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■ Renew bolts/nuts

Torque converter

Engine/gearbox attaching bolts:

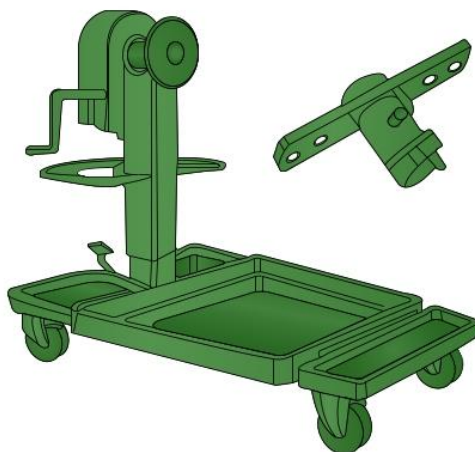
M8: (19 Nm)

M12: (66 Nm)

Transfer box to gearbox: (56 Nm)

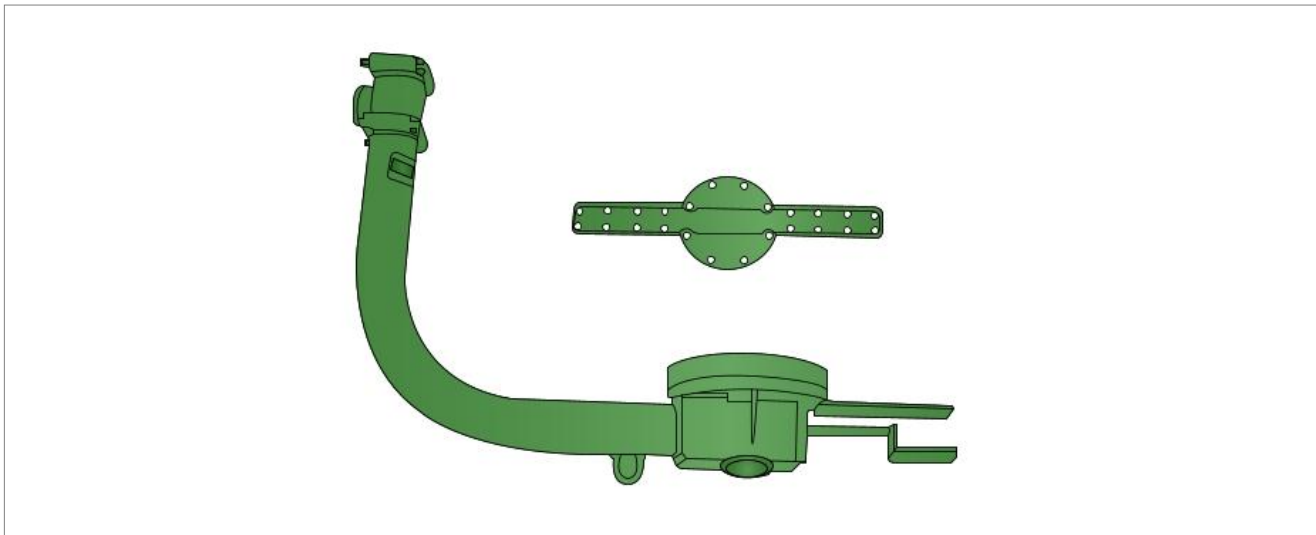
Special tools

Engine support: (00 2 300)



Engine support: (00 2 320)

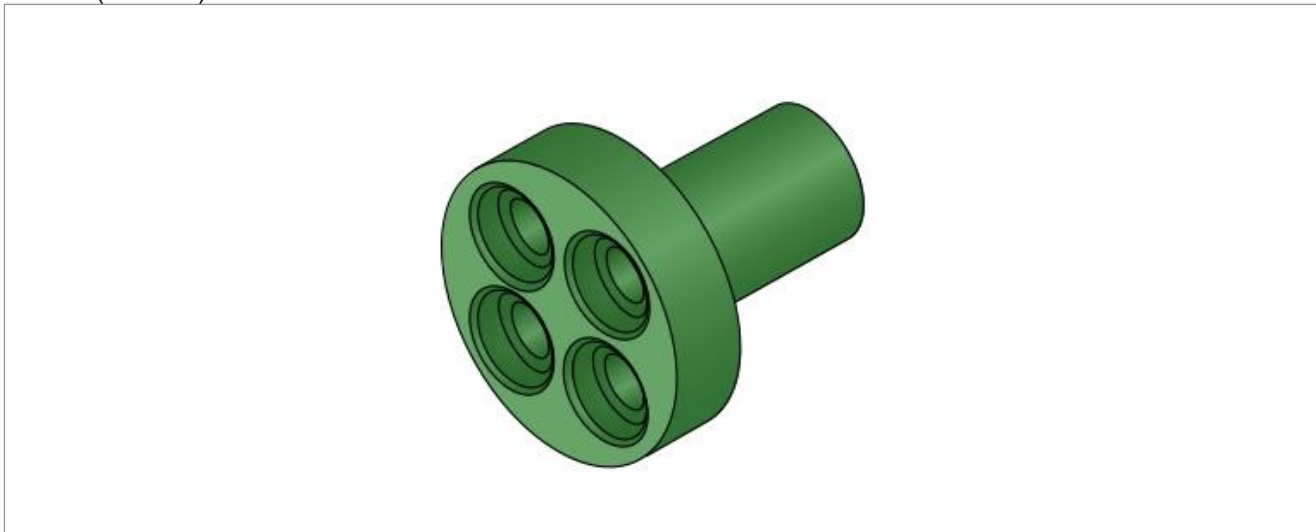
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Lifting tool: (11 0 070)

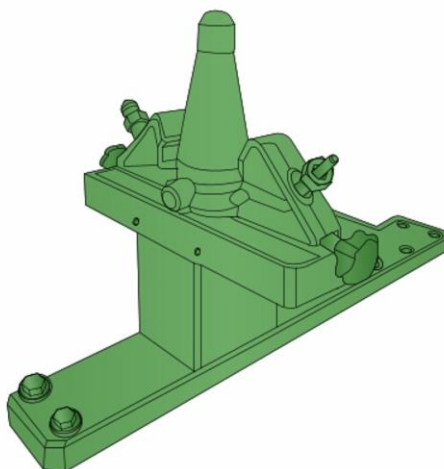


Socket: (11 6 480)

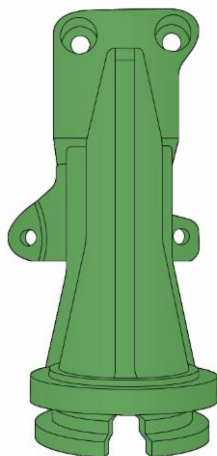


Engine support: (2 413 082)

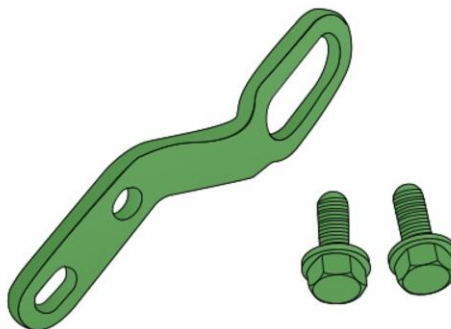
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Engine support tool: (2 455 010)

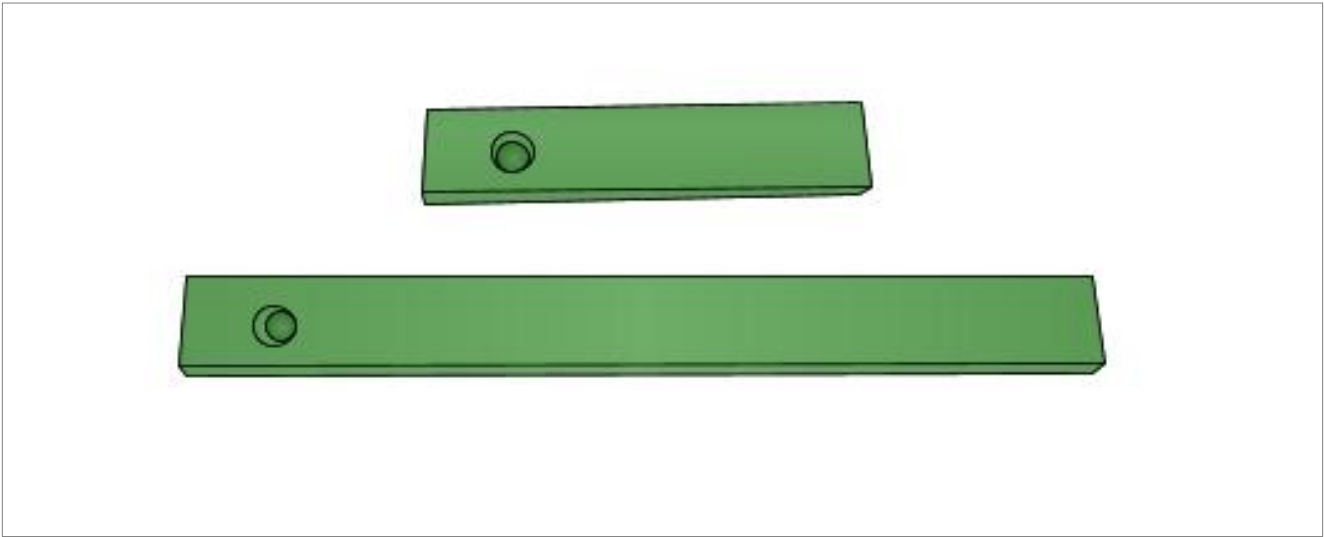


Lifting eye: (2 459 012)

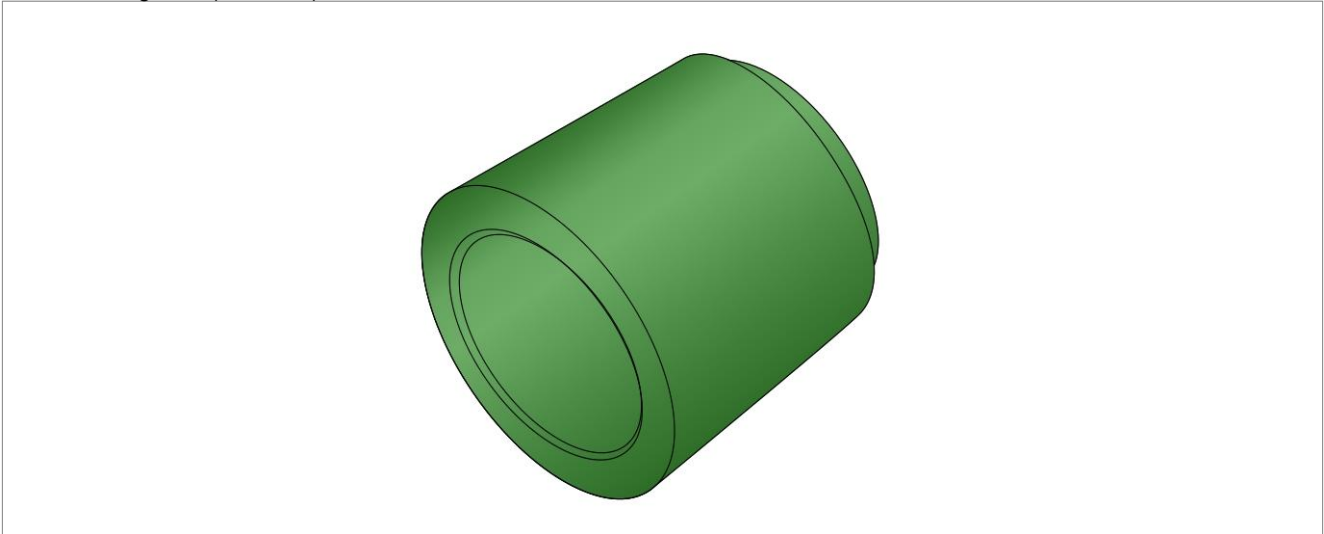


Special tools: (11 8 780)

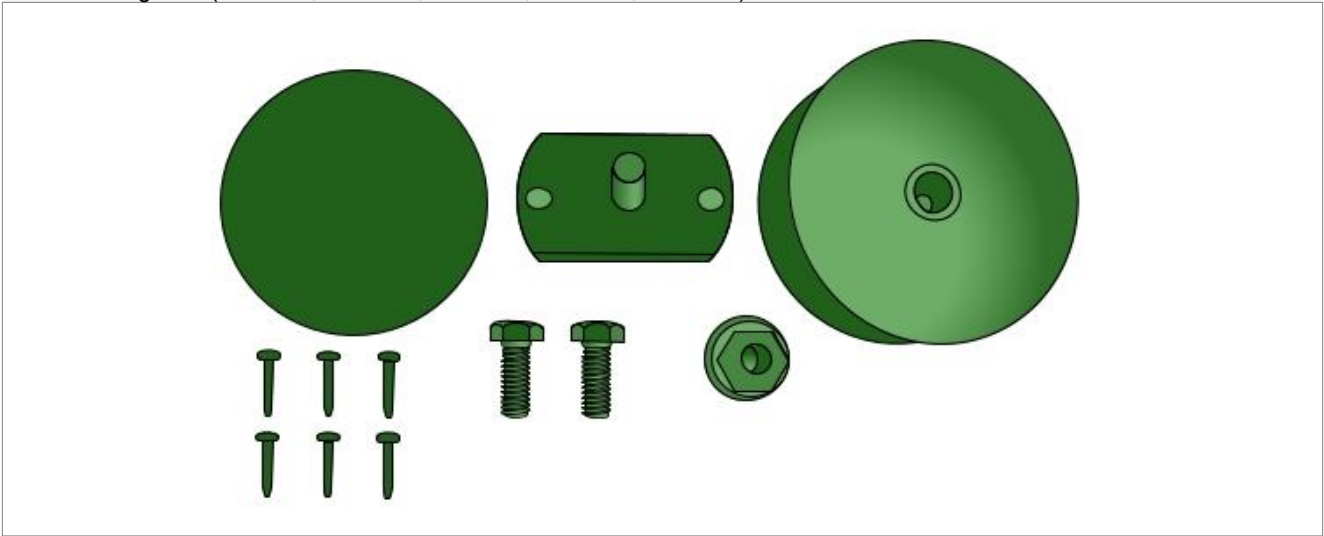
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Oil seal fitting tool: (11 8 815)

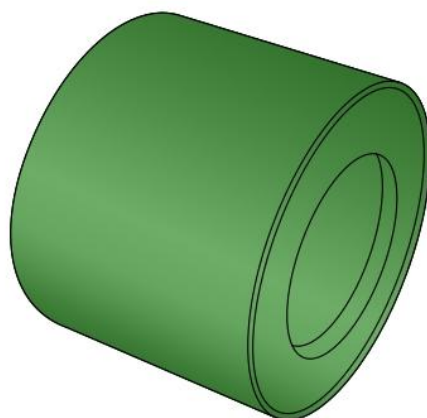


Oil seal fitting tool: (11 9 181, 11 9 182, 11 9 183, 11 9 184, 11 9 185)

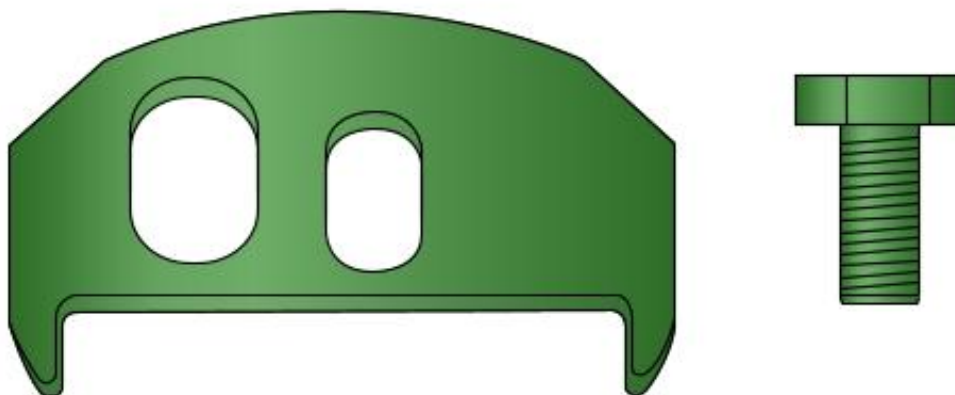


Bush: (11 9 265)

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Flywheel locking tool: (11 9 811)



Adaptor: (2 360 215)

